

Question Number	Answer	Additional Guidance	Mark
1(a)	An explanation that includes the following points: • because (antioxidants / vitamin C) reduce (the quantity of) free radicals (1) • that cause damage to {cells / tissues / blood vessels / endothelium} (1) • reducing {plaque / atheroma} formation (1)	Accept neutralise, get rid of, inhibit, plus many other ways of saying this Ignore ref to clots	Expert (3)

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	A description that includes the following points: • use {equal / controlled} masses of the foods (1) • use {standard / same} extraction method (1) • measure out {equal / same} volume of DCPIP solution (1) • add food extract dropwise until {no colour remains / blue to colourless} (1) • record {volume / number of drops} used (1) • explanation of how to calculate vitamin C content / description of calibration of DCPIP with known vitamin C concentration (1)	Accept stated detail e.g. volume of distilled water / time of macerating / standard filtering method measure out {equal / same} food extract Add DCPIP solution dropwise until stays blue / colour stays / no change e.g. $\text{conc. vitamin C} = \frac{\text{vol. of standard Soln. vit C}}{\text{volume of food extract added}}$ Ignore colorimeter, colour charts	Expert (5)

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	types of {food / fruit}	ALLOW (storage) time / (storage) temperature	Grad (1)

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	An answer that includes the following steps: - concentration at start – concentration after 3 weeks (1) - correct answer to mp1 divided by 3 given to 2 significant figures (1)	correct answer with no working gains both marks e.g. $4.4 - 1.6 = 2.8$ e.g. $2.8 \div 3 = 0.93$, accept .93 ignore sign, 0.93333 would get 1 mark	Expert (2)

Question Number	Answer	Additional Guidance	Mark
1(c)(ii)	An answer that includes the following points: • increase in storage time decreases the vitamin C concentration (in all juices at both temperatures) (1) • vitamin C concentration is less at 4°C (than 20°C) in orange and lime (1) • vitamin C concentration is more at 4°C (than 20°C) in (carrot and) pineapple (1) • very little difference in effect at 4°C and 20°C in carrot (1) • no {range bars / standard deviation} so cannot tell if differences are significant (1)	Piece together vitamin C concentration more affected at 4°C (than 20°C) in orange and lime vitamin C concentration less affected at 4°C (than 20°C) in (carrot and) pineapple (1)	Expert (4)